

Primer

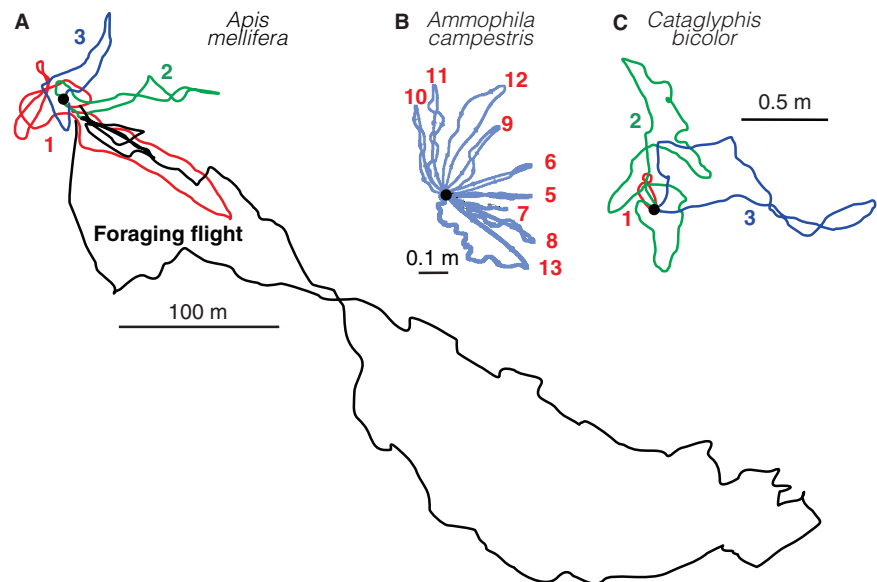
Insect learning flights and walks

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Knowledge of where things are in one's habitual surroundings cannot be encoded genetically and must be acquired in those surroundings. Many ants, bees and wasps forage from a home base and before doing so learn where resources are to be found and how to return with them to their nest. A significant component of this navigational learning seems to be the acquisition of panoramic views that insects record close to their nests and along the paths between these places. Behavioural evidence indicates that these views are retinotopic, meaning, for instance, that an insect knows that it faces along a familiar route, if the image on its retina matches a view that it had previously recorded, when facing in that direction during route learning.

This learning is not the result of a reward but is rather a kind of 'genetically encoded anticipatory learning'. It is analogous in some ways to how young indigo buntings, before they migrate, learn the position of the North Star in the sky. The bird's method is to memorise the pattern of stars that they see rotating about a fixed star, anticipating what they will need to know to control their migratory direction. Finding a location within variable terrain is harder and an insect's anticipatory learning is correspondingly complex. The behaviour underpinning learning is particularly striking when the insects first leave their nest and it can tell us much about spatial learning on a variety of scales.

In what follows, we first consider what is now understood about bee learning flights at a large scale. Study of these flights has revealed some of the strategies that bees employ to become familiar with a foraging area. We then turn to the components of learning flights and walks that occur in the immediate vicinity of the nest. At this finer scale, it is possible to record the exact manoeuvres during which insects learn about visual features



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Figure 1. Initial excursions of insects exploring different compass directions around the nest. (A) Successive orientation flights of a honey bee before a first foraging flight. Nest marked by (•) (modified from Degen, J. *et al.*, 2015). (B) Short flights of an *Ammophila campestris* wasp while digging a new nest (•). Numbers indicate the sequence of successive flights (modified from Baerends, G.P. (1941)). (C) Successive orientation runs of a *Cataglyphis bicolor* ant on first appearing outside the nest (modified from Wehner, R., *et al.*, 2004).

near their nest or a feeding site. With such detail one can postulate when learning occurs, what is learnt and how learning is embedded within stereotyped manoeuvres. Moreover, by recording return trips to the nest one can see close coordination between movements underlying learning and its usage. Many animals face similar problems, but the retinotopic nature of scene learning in insects, combined with their stereotyped behaviour, makes the insects' solutions especially obvious.

Large scale exploration

Most of us adopt a similar strategy when becoming familiar with a new place, as pithily expressed by an Australian aborigine: "I don't go far in the beginning, I go some distance and come back again, then in another direction and come back, and then again in another direction. Gradually I know how everything is and can go out far without losing my way" (quoted in Harold Gatty's 1999 book *Finding Your Way without Map or Compass*). This recipe can be applied both to wild and urban environments and in both locations the process is much easier

with a good view of the sky and large landscape features, be they mountains or monuments.

After a flight close to their nest, sometimes lasting 20 seconds or more, bees circle higher and survey a wider area. The latter part of a honeybee's and bumblebee's learning flights, also called orientation or exploration flights, could only be observed over short distances until the recent development of radar tracking using a transponder attached to the bee to generate a recognisable signature. The bees' strategy, as shown with this technique, resembles that of humans: the area covered by a honeybee or bumblebee increases over successive flights as the bee explores a widening range of distances and directions (Figure 1A).

A honeybee's spatial knowledge after a single learning flight has been tested by catching and displacing the bee when it leaves the hive for the second time. In the first such study, bees released 250 m from the hive took only a few minutes to reach home, whereas, with 500 m release sites, homing took much longer. Monitoring departure directions on release revealed that bees flew directly

homeward provided that they could see the immediate surrounds of the hive from the release point. With radar tracking, bees were shown to home rapidly and directly from release points within the narrow area explored during their first orientation flights, but they engaged in search when released in unexplored regions. Bees in a single learning flight can, thus, learn enough about the terrain that they have explored to return swiftly home.

Young honeybees on first leaving their natal hive are untried navigators, ignorant of their surroundings. Perhaps, for this reason, they are cautious in their orientation flight and tend to move straight out and then return straight back to the hive (Figure 1A). This hair-pin flight pattern has several advantages for the inexperienced: it might be useful for calibrating other navigational mechanisms like path integration; it enables a visually guided return leg that could reinforce the learning of views that were stored on the outward leg of the path. The second and third orientation flights are similar, but longer and in different directions. On a much smaller scale, the brief flights of the wasp *Ammophila* while digging a burrow in the ground and the learning walks of ants replicate the findings with bees: the insects successively sample different directions and range further and further from the nest (Figure 1B,C).

Currently, radar just gives a bee's X–Y position every three seconds and does not reveal the details of its learning manoeuvres. Examination of the start of learning flights or walks, with the insects close to the nest, has the advantage that the insect's movements can be recorded with video at a high temporal and spatial resolution. One can then monitor the directions in which the insects face and from where they look towards the nest entrance. This information indicates which views are likely to be memorised and how the learning manoeuvres may be adapted to acquiring information in a form that is suited to guiding efficient returns to the nest.

Learning manoeuvres close to the nest

The initial discovery of learning flights was due to the skilled observation and insights of past generations of

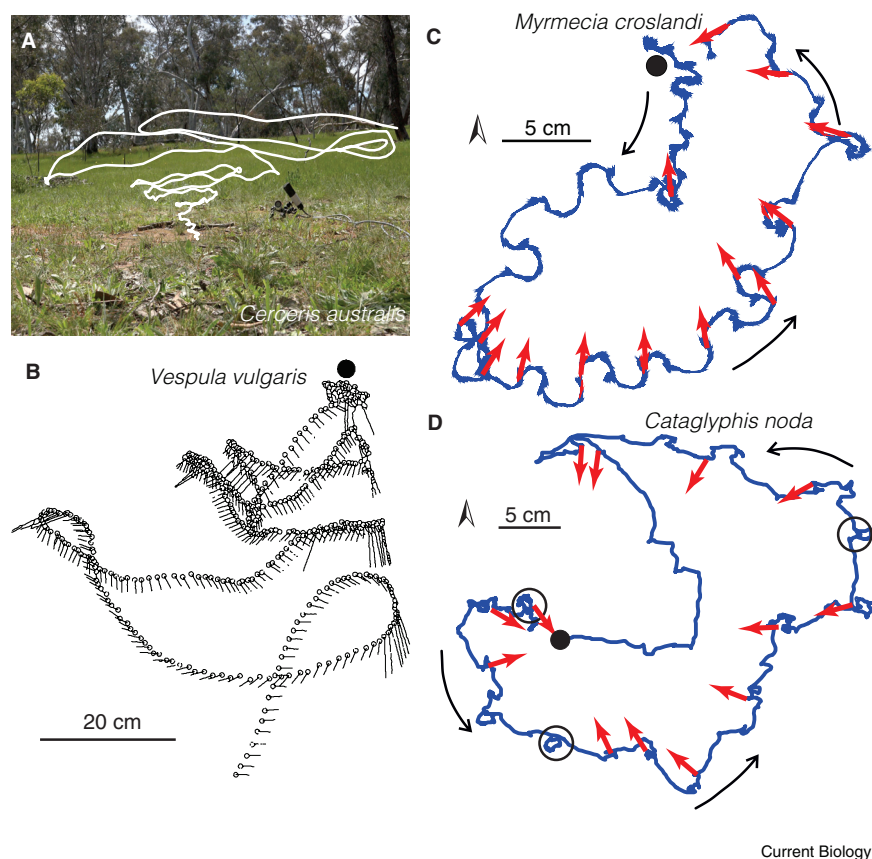


Figure 2. Learning manoeuvres close to the nest.

(A) The learning flight of a ground-nesting wasp (*Cerceris australis*) as seen from the side. (B) Top-down view of learning path and body orientation of a social wasp (*Vespula vulgaris*). Longer glyphs mark when wasp looks at nest (modified from Collett and Lehrer, 1993). (C,D) Two examples of learning walks in ants. Red arrows show where ant turns to look back at the nest (●). Circles in (D) mark where ant turns through 360°. (C) Unpublished data of Jayatilaka, P., Narendra, A., Murray, T., and Zeil, J. (ms in preparation). (D) Modified from Fleischmann, P.N., et al., 2017).

naturalists. The brief account by Bates in 1863 of the wasp *Bembex ciliata* leaving its nest seems to be the earliest that is currently known, though Bates had read of “Something analogous in hive bees”: “After making a gallery (in its nest) the owner backs out and takes a few turns around the orifice... On rising in the air.... the insects flew round over the place before making straight off.... I was convinced that the insects noted the [locality and] bearings of their nests and the direction that they took in flying from them.... [it] seems to be a mental act of the same nature as that which takes place in ourselves.... (T)he senses, however, must be immeasurably more keen and the mental operations more certain... than in man for to my eye there was no landmark on the even surface of the sand... and the borders

of the forest ... were not nearer than half a mile.”

Learning manoeuvres close to the nest have features that are common across different species, whether they are walking or flying (Figure 2). Indeed, ants evolved from wasps and the strong resemblance between wasp and ant ‘turn-back-and-look’ behaviour when learning their nest surroundings encourages the notion that ant learning walks are derived directly from wasp learning flight behaviour (Figure 3). The most significant property common to learning flights and walks is that, in both cases, the insects look back to face the nest while moving in arcs or complete circles around it and thereby experience views across the nest from widely distributed vantage points.

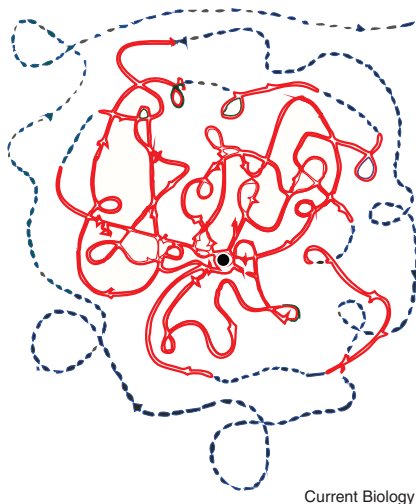


Figure 3. The wasp, *Astata unicolor*, walks and flies during its locality study.

Red: walking, blue: flying (recoloured from Peckham, G.W. and Peckham, E.G., 1905).

The efficacy of these learning manoeuvres is illustrated by Tinbergen's seminal experiment of almost a century ago. A ring of pine cones was placed around a sand wasp's nest hole when the wasp was inside. After the wasp had emerged, performed a learning flight, and departed, the ring was displaced 30 cm from the nest. On its return, the wasp flew straight to the location indicated by the pine cones, ignoring the real nest site and the cues that had previously marked its location. The most recent learning flight had generated new memories that overrode the old ones, but only those specifying the nest itself; the new memories worked seamlessly with those that brought the wasp to the nest's general area.

More is now known about the way that stored retinotopic views are likely to guide homing. Ants placed on a familiar visual route will turn until they are facing in the route direction and then continue in that direction scanning back and forth to check the view. Essentially, the ants are matching or aligning their current view to a stored one. No knowledge of distance is needed for keeping to a direction. Indeed, simulations with natural scenes show that alignment matching can be achieved using unsegmented panoramic views. Although the same could hold for specifying locations, in

fact wasps and bees use optic flow to segment the scene and assess the distance of salient features near the nest that can best pinpoint its position.

In one experiment that demonstrates the likely use of motion parallax in estimating object distance, *Cerceris* wasps became accustomed to a single cylinder placed close to their nest. A wasp on its return was confronted with a cylinder of a different size moved further from the nest: the wasp searched for its nest at a point defined by the distance between the cylinder and the nest during learning rather than by the expected image size of the cylinder. Analogous experiments on honeybees and ground-nesting bees gave similar results. The importance of this finding is twofold: it emphasises that learning flights are concerned with more than the acquisition of static views and it points to the intriguing question of what constitutes the memory of object distance specified by motion-parallax.

Patterns of learning flights and walks

The visual scenes around the nests of insects can differ dramatically, depending on the species' habitat. The nest could be in a tree or in bare ground or in undergrowth or in a featureless salt-pan and there is much to be discovered about whether and how an insect's learning choreography is adapted to these different locations. A *Cerceris* wasp, which nests in open ground, tends to leave its nest in a sequence of expanding arcs of increasing height (Figure 2A), pivoting alternately clockwise and counter-clockwise about the nest, flying obliquely to the orientation of its body so keeping the nest and its surrounds in the fronto-lateral field of one or the other eye. It faces the nest directly, but very briefly, at the end of each arc when it turns and switches its direction of rotation.

Similar expanding arcs are found in other species, in which fixations of the goal at the end of arcs can be quite prominent (Figure 2B). The same diversity occurs in ant learning walks. During their circuit around the nest desert ants periodically perform full rotations on the spot. Some species of desert ants spend longer facing in the nest direction than elsewhere, but

others, such as Australian bull ants, do not. The ants' nest-directed turns are in *Cataglyphis noda*, at least, guided by path integration, with compass direction provided by the Earth's magnetic field.

Experiments with bumblebees and ground-nesting wasps indicate that the radial position of learning flight arcs relative to the nest is flexible and depends on the distribution of objects close to it. If a cylinder is placed a few cm from a nest in otherwise open surroundings, the arcs occur in roughly the opposite radial position so that the wasp faces the cylinder when looking across the nest (Figure 4). The significance of this feature becomes clear when subsequent return flights are analysed. The insects, on their return, approach the cylinder and nest in the same compass direction in which they backed away from them during their learning flight (Figure 4). Displacement of the cylinder does not influence where the insect searches relative to the cylinder. This test indicates that the insects must learn the compass direction in which they back away. Matching facing directions of returns to those of learning flights enhances the probability of encountering stored views. Additionally, the choice of direction means that the insect can be guided by dominant visual features all the way to the nest.

Returning wasps and bees vary in how they make use of the nest facing views that are potentially memorised during learning flights. For instance, a *Cerceris* wasp flying towards home enters the cone of nest directed views (Figure 2A) and, on encountering a view, turns in the nest direction. A bumblebee's zigzag flight path on returns matches the spatio-temporal pattern of the loops of its learning flights such that returning insects tend to face the nest at the bends of the zigzags and at equivalent points in loops.

In both insects, similarities between learning and return flights extend to detailed patterns of motion so generating similar patterns of optic flow between memory acquisition and use. As already mentioned, tests to demonstrate the likely use of motion parallax to estimate the distance of visual features from the nest have

to control for the use of image size as an alternative cue. Under normal circumstances the two cues will reinforce each other, so that the usual operation could be the linkage of a view with an optic flow signature.

When is an insect's locale learning complete?

Although insects can home after a single learning flight, their learning manoeuvres continue over several subsequent departures. Learning also occurs during approaches to the nest when an insect can validate and adjust what it has learnt. If a returning insect has difficulties in locating its nest, either because the scene has changed during its absence or because the insect has not learnt enough for efficient homing, the next learning flight is abnormally long. Failure or extended search for the goal thus automatically triggers extra learning (with no need for parental encouragement).

In honeybees and bumblebees, learning on departure tends to continue over half a dozen or so trips. Both departures and returns become shorter and less elaborate during this sequence of flights. With simplification of the path, honeybees and *Cerceris* wasps, in a kind of primitive skill learning, dispense with motion parallax and optic flow and rely instead on two-dimensional image matching for guidance, searching for the nest where a nearby visual object has the same image size and retinal position as it did when acquiring views close to the nest.

Learning flights and walks do not only occur when an insect leaves its nest. Insects also perform them on leaving a newly discovered food location. And a parasitic wasp performs them on observing its host digging a nest. It can then return to inspect the nest site and, when the moment is right, deposit an egg. Details of learning manoeuvres may vary to suit the particularities of these places. Learning an inconspicuous nest entrance is more demanding and more crucial than learning the location of a brightly coloured and scented flower. These differences are reflected in the length of learning flights and the number of successive departures accompanied by a learning flight. Both are longer when bumblebees

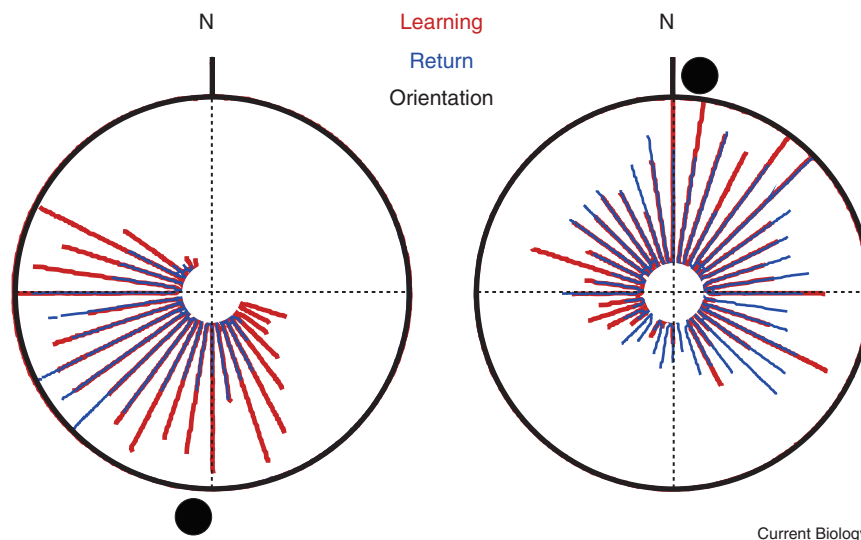


Figure 4. Wasps' similar facing directions during learning and homing are set by nearby visual features.

Circular histograms of body axis orientation during a learning (red) and return flight (blue) of two wasps. For one wasp (left) a cylindrical landmark was south of the nest and for the other (right) it was north (replotted from Zeil, J., 1993).

leave their nest than when they leave a flower, even if the flower and nest hole and the surrounds of each are made to look similar. Thus, the type of destination may set the insect's investment in learning, possibly in terms of the number and the spacing of recorded views.

What next?

The study of learning flights and walks suggests that learning occurs primarily during moments when insects look back to face the nest and that it could be fruitful to focus behavioural and physiological studies on these moments. There are many questions. For instance, are recorded views snapshots or videos? How are views linked to celestial compass cues? And how are views, image motion patterns, and compass information processed, stored and integrated in the insect brain? A start has been made in identifying central brain regions in ants and honeybees that are activated and modified as a consequence of an insect performing learning walks or learning flights.

Detailed studies of the central complex in several insects show how orientation can be controlled by both celestial compass and landmark cues. It may soon be possible to investigate physiologically

the navigational instructions that a homing insect extracts from its comparison of memorized views and the current scene and how it generates efficient behavioural output. Much is likely to be learnt from experimental approaches now being used in which neural activity is recorded from tethered insects flying or walking in virtual reality.

Finally, we still know too little about differences between the learning behaviour of species that have different modes of locomotion and different habits and habitats. The first wasp watchers were entranced by the diversity they found, but they could only sketch and recount their observations. With current video techniques and precise ways of mapping the visual surroundings in which a species lives, it is time to revisit these insects (if they are still around) and extract how learning and homing behaviour are adapted to the specific problems they face. To give a flavour of early observations: the bembecine wasp *Bicyrtes fasciata* returns to her nest in a semi-barren sandy area "on the wing, flying high in the air, and poises in mid-air ten or twelve feet above the hole, then drops straight down as if with a parachute, digs for a moment at her very feet where she alights, and lo! her burrow

opens". What learning flight and sequence of recorded views underpin such precise targeting?

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Primer

Neuroethology of spatial cognition

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A key challenge for animals is recognizing locations and navigating between them. These capacities are varied: we can remember where our car is parked at the mall, rats are able to remember where their nest location is while foraging for food morsels, and bats are able to fly directly to a favourite fruit tree 20 kilometers from their home cave. These spatial abilities, whether commonplace or remarkable, raise fundamental questions. First, how do animals find their way? Second, how does the brain represent the outside world?

In this Primer, we will address both questions from the perspective of rodent cognition and neuroscience.

Rodent behavior basics

Rodents, particularly rats and mice, are much studied as model organisms for understanding basic mammalian brain processes and, increasingly, disease states. At the beginning of the 20th century, rats were used to answer basic questions about learning and memory, often using mazes. Much is now known about their behavior and neural systems, but it is useful to consider their natural behaviour – or at least the unstructured behaviour of a laboratory rat foraging for food – on an open platform.

Exploration

Rats are cautiously curious. If you were to place a rat on a table, in the absence of strong light (which they

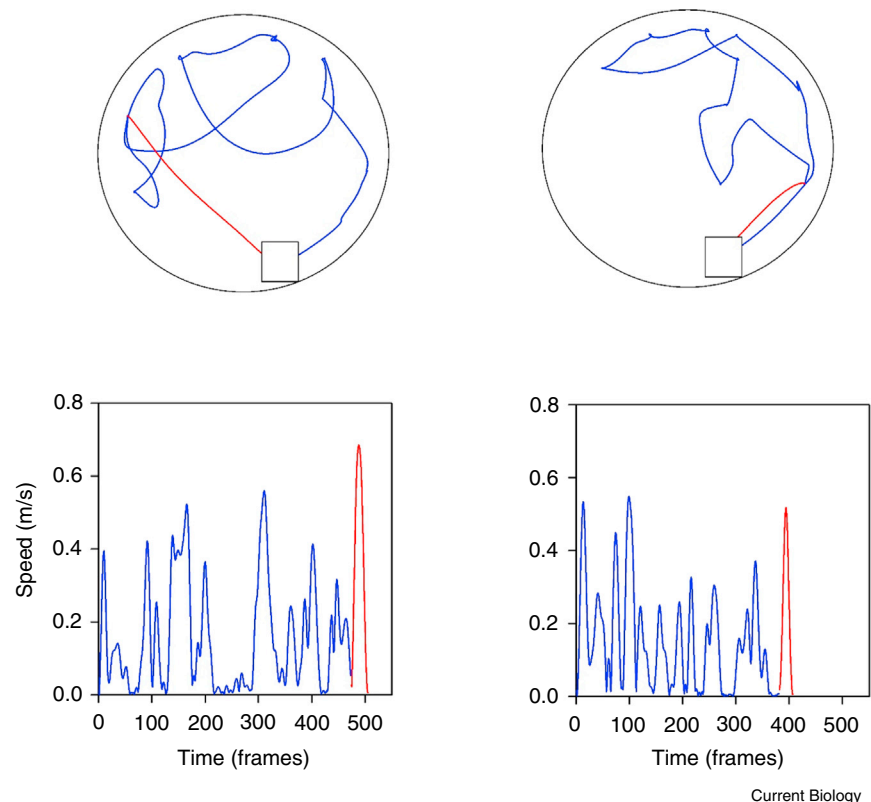


Figure 1. Topographic (top panels) and kinematic (bottom panels) characteristics plots for two representative exploratory trips by a rat.

The outward segments (blue lines) are a set of slow, relatively non-circuitous progressions punctuated by stops of varying duration. The homeward segment (red lines) is a single progression with peak speed at the midpoint of the journey and magnitude scaled to the Euclidean distance to the home base. Longer homeward segment Euclidean distances elicit faster peak speeds (left panels), whereas shorter homeward segment Euclidean distance elicit slower peak speeds (right panels).